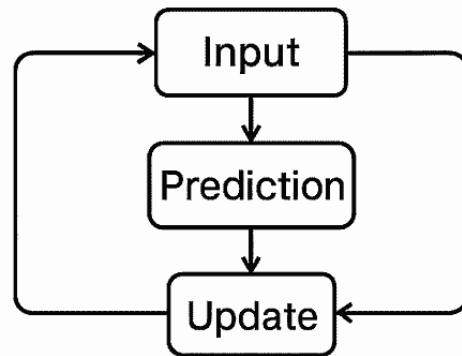
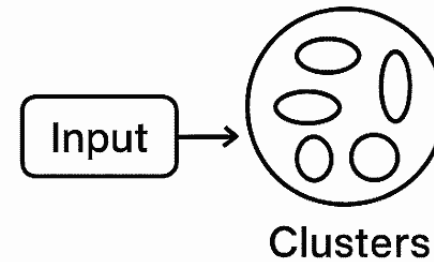


Bird's Eye View of Reinforcement Learning

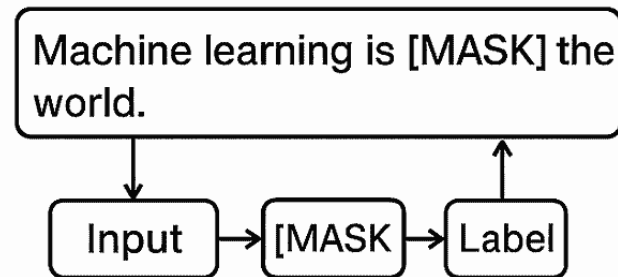
Supervised Learning



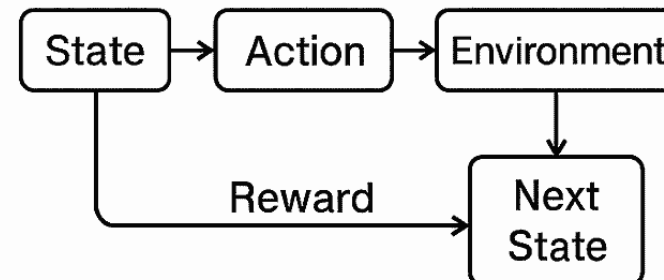
Unsupervised Learning



Self-Supervised Learning



Reinforcement Learning



Reinforcement Learning from Human Feedback (RLHF)

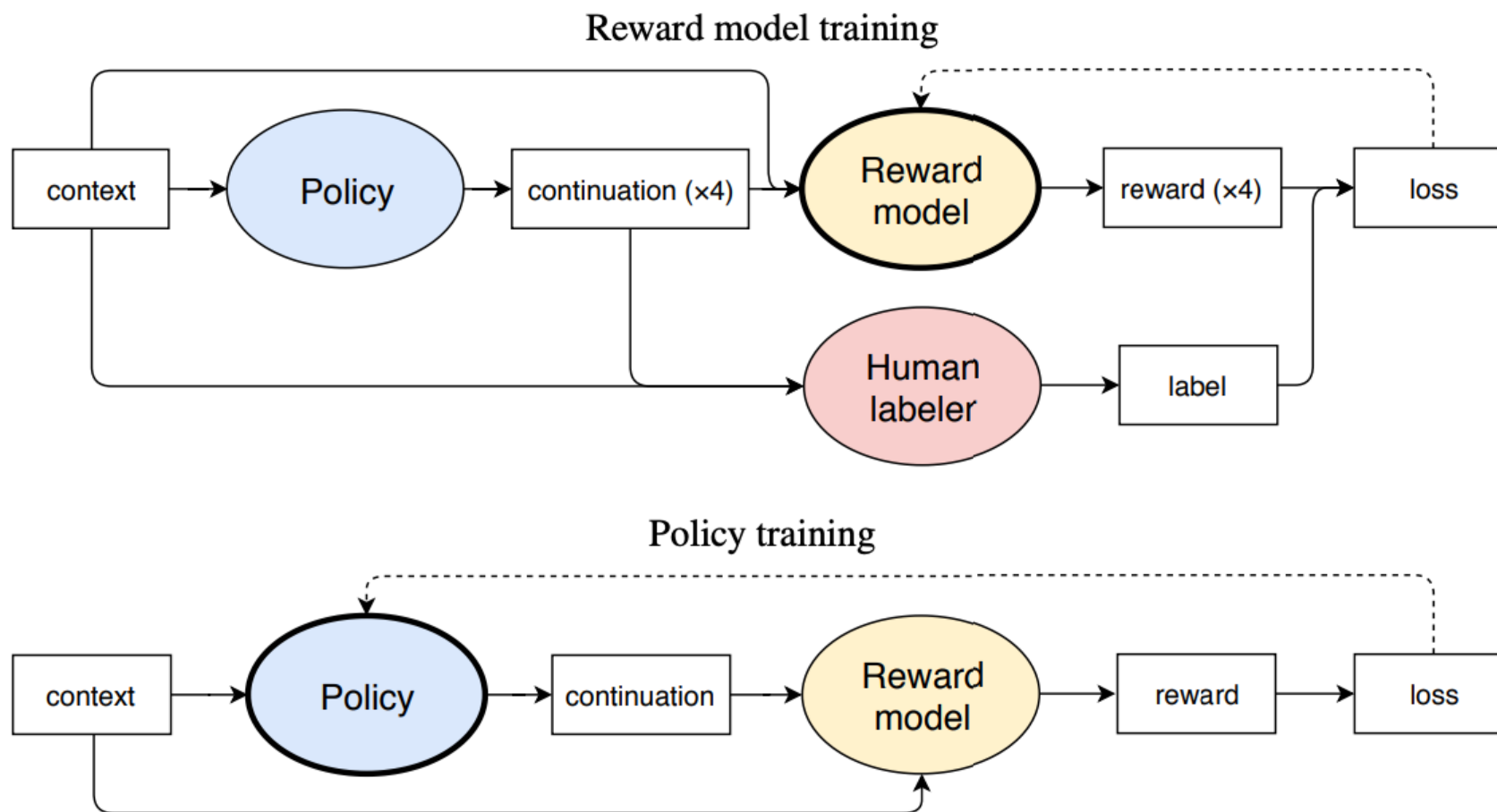
- Seeking to address the exposure bias and to leverage negative examples, we define a language model $\pi(\cdot|x)$, where x is some prefix, by initializing it from the SFT model
 - π is called the “policy”
- We assume that there is some real function $r(x,y)$ where y is a continuation of x , that scores the different continuations, such that the better y is, the higher r is
- We seek to optimize the expected reward:

$$\mathbb{E}_{\pi} [r] = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi(\cdot|x)} [r(x, y)]$$

Reinforcement Learning from Human Feedback (RLHF)

- The difficulty is that r is unknown to us, so we need to train it as well
- We assume that we have a sample of prefixes x (instructions) and for each, we have 4 options, sampled from our current LM p . We further assume annotation of which of the 4 options is best.
- We train p to increase the expected reward by changing its probability. In parallel, we train r to conform to the preferences of the humans.
- A penalty term keeps π from diverging too much from the SFT LM

Reinforcement Learning from Human Feedback (RLHF)



Results

- As mentioned last class: very difficult to evaluate
 - Many benchmarks used, testing for a wide variety of capabilities

Training for MATH does not transfer well to Deepmind Math. One difference between MATH, which is part of our development evaluations and DeepMind Math, which is in our unseen evaluations is that the former often

Models generally overfit to IFEval. We find that there is a significant difference between performance on IFEval and IFEval-OOD of all the models, even though we created the latter to be structured very similar to the original dataset, just with a disjoint set of constraints. We observe that instruction following with verifiable constraints is a

Instruction following performance varies across categories. We observe that the relative performance of TüLU 3 models on AlpacaEval is different from that on HREF. This may be explained by the fact that instruction following is a highly diverse task, and the distributions of HREF and AlpacaEval may differ, with some categories of instructions